Nouh Taha CHEBCHOUB

AI Engineer & Data Scientist

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Profile

AI Engineer and Data Scientist specialising in physics-informed ML, LLM automation, and end-to-end MLOps. Published researcher (Sensors, Springer) with production experience across EV digital twins, real-time telemetry, and intelligent document processing. Open to international opportunities.

Professional Experience

AI Engineer — *Variable Data Co.* July 2025 – Present

- Designed an end-to-end LLM-based engine to extract and cross-compare clauses across Medical, Motor, and General insurance quotations, producing structured analysis sheets for client decision-making.
- Reduced report turnaround from **2+ days to under 30 minutes** via optimised prompt-engineering and structured-output pipelines — maintaining human-level accuracy across all insurance categories.

Data Scientist Intern — *DHBW Friedrichshafen, Germany* Feb 2025 – July 2025

- Deployed a Digital Twin for an EV in CARLA, integrating real-time telemetry at **10k+ data points/sec** with **<50 ms latency** for predictive analytics and proactive maintenance.
- Architected a PINN for Battery SoC estimation by embedding the Coulomb Counting ODE into the loss function, achieving **<2% prediction error** with guaranteed physical consistency.

Selected Projects

TwinFlux — End-to-End EV Energy Intelligence Platform

- Physics-calibrated EV simulation (SUMO + HVAC) achieving **±0.6% Wh/km** across 5 vehicle profiles; AutoML pipeline (CASH + Physics-Informed NAS, Optuna) feeds a real-time SoC prediction API with MLflow tracking. *Stack: Python, FastAPI, PyTorch, PostgreSQL, MinIO, MLflow, Docker.*

EEG-Based Emotion Detection — DEAP Dataset *(Published · Sensors, MDPI)*

- CNN & RNN (GRU, Bi-LSTM) models on DEAP (32 participants, 1,280 trials) reaching **91–94% accuracy** in valence/arousal classification; findings published in *Sensors*, MDPI — doi:10.3390/s25061827.

GAN-Sim — Generative Behavioural Modelling for Agent Training

- Conditional GAN trained on 5,000+ annotated frames to generate synthetic agent trajectories for RL environment augmentation; achieved **93% target-detection accuracy**, demonstrating sim-to-real behavioural transfer.

Education

M.Sc. Data Science — Faculty of Science Semlalia, Cadi Ayyad University, Marrakech Sept 2023 – July 2025

B.Sc. Mathematics & Computer Science — Faculty of Science Semlalia, Cadi Ayyad University, Marrakech Sept 2020 – July 2023

Technical Skills

ML / AI: PyTorch, TensorFlow/Keras, HuggingFace Transformers, LLMs, Computer Vision (YOLO/OpenCV), GANs, PINNs, NLP, XAI

MLOps & Data: MLflow, Airflow, FastAPI, Docker, PostgreSQL, MongoDB, InfluxDB, MinIO, Grafana

Languages: Python, SQL **Spoken:** English (fluent), French (fluent), Arabic (native)

Publications

- **Enhancing EEG-based emotion detection with hybrid models** — *Sensors*, MDPI, 2025. doi:10.3390/s25061827
- **Fostering Innovative Learning Through Augmented Reality** — *LNNS*, Springer, 2025. doi:10.1007/978-3-031-94623-3_37

Certifications

- **Machine Learning Specialisation** — DeepLearning.AI **Python for Data Science, AI & Development** — IBM